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CRJU 705

Final Project Report

Introduction

According to the social disorganization theory, locations with higher levels of poverty, minority populations, and urbanization are more likely to experience higher crime rates. As it suggests in the name, these locations experience more crime because they are less organized as a community. They lack support and self-sufficiency, which results in them adopting criminal behaviors and tendencies. For this project, I attempt to see if murder rates in the United States can be explained by social disorganization. Specifically, the question I aim to answer is: How does poverty, unemployment, and living in an urban area impact murder rates? Based on the theory of social disorganization, my hypothesis is that all three variables will increase murder rates.

Methods

For this project, I utilized data from the crimestatewide_tbl_df dataset in R Studio. After cleaning and filtering out unneeded variables, I was left with a dataset named crime2. The crime2 dataset includes murder rates, unemployment rates, poverty rates, and the percentage of the population living in a metropolitan area for all 51 U.S. states (D.C. is its own state which makes 51 instead of 50). All variables are measured as percentages. All of the data cleaning and analyzing was done in R. To test my hypothesis, I conducted correlation tests and used multiple linear regression to find the impact of the three independent variables (Poverty, unemployment, and metropolitan living) on murder rates (the dependent variable). For both analysis and interpretation, I used a Frequentist approach. Because of this, the results are also tested against

the null hypothesis, which assumes that there is no significant relationship between the independent and dependent variables.

Results

Table 1 shows the associations of each of the three independent variables on murder rates. It shows the correlation coefficients, 95% confidence intervals, and p-values for each variable. For all three, a low to moderate positive association can be seen. Poverty has a correlation of .42, unemployment rate has a correlation of .39, and living in a metropolitan area has a correlation of .33. Based on the Frequentist approach, it can be said that since the p-values for all three are below .05, there is enough evidence to reject the null hypothesis, which means that there is strong confidence that these correlations are statistically significant.

Variable	Correlation Coefficient	95% CI	p-value
Poverty rate	+ .42	.17 - .63	.0018
Unemployment rate	+ .39	.13 - .60	.0045
Living in a metropolitan area	+ .33	.06 - .56	.017

TABLE 1. results from Frequentist correlation test

Table 2 shows a Frequentist multiple linear regression model. For every 1% increase in poverty rate, murder rates increase by about .7, holding the other two variables constant. For every 1% increase in unemployment rate, murder rates increase by about 1.11, holding the other two variables constant. Finally, for every 1% increase in the population living in a metropolitan area, murder rates increase by about .11, holding the other two variables constant. The effects of poverty rates and living in a metropolitan area were the only statistically significant outcomes, with p-values of less than .05. Visual representations of these results can be found in figures 1, 2, and 3.

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.54303    3.94284  -3.688 0.000584 ***
poverty       0.70663    0.27748   2.547 0.014211 *
unemployed    1.10195    0.85329   1.291 0.202877
metropolitan  0.10777    0.03267   3.299 0.001857 **
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

TABLE 2. Frequentist multiple linear regression model

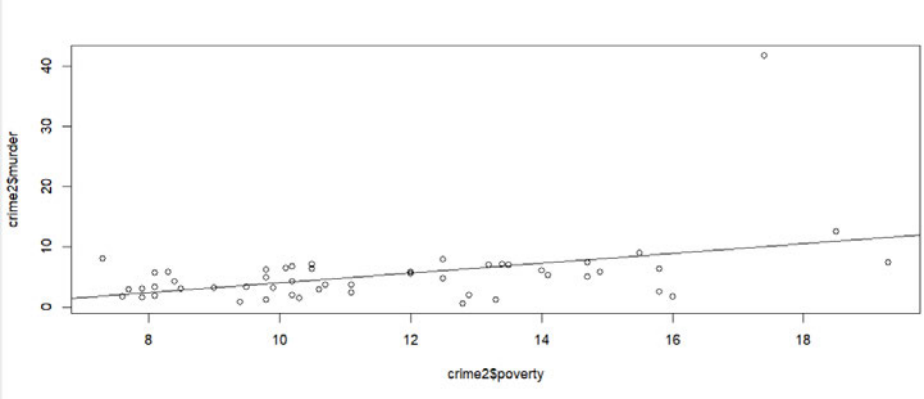


FIGURE 1. Scatterplot poverty and murder rates with regression line

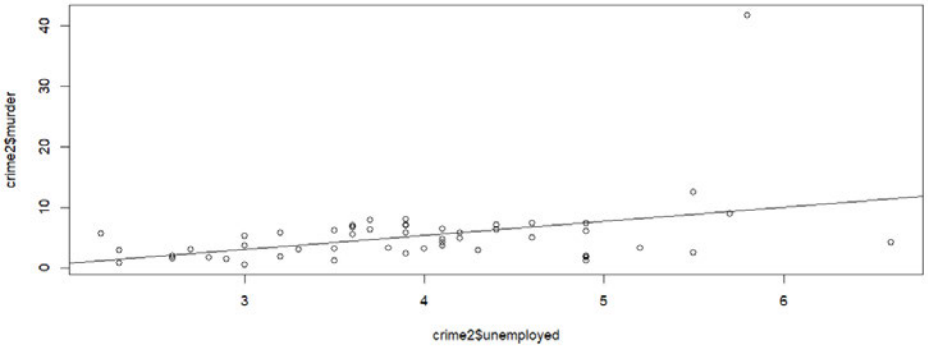


FIGURE 2. Scatterplot of unemployment and murder rates with regression line

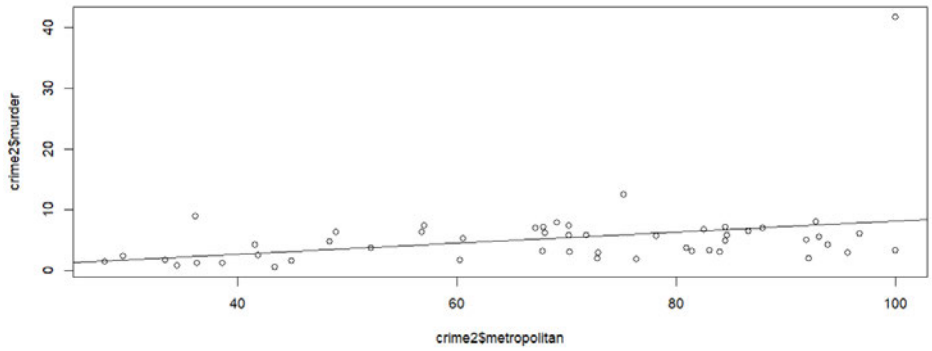


FIGURE 3. Scatterplot of living in a metropolitan area and murder rates with a regression line

Discussion

The results of the regression model partially support my hypothesis that all three independent variables would increase murder rates. The effects of poverty and metropolitan living were found to be statistically significant, but the effect of unemployment rates on murder rates was not. There was a positive correlation between each of the variables and murder rates, which also supports my hypothesis. Additionally, p-values of less than .05 allowed me to have enough evidence to reject the null hypothesis. While nothing is 100% certain, I have high confidence that each of the variables are positively correlated with murder rates. I also have confidence that poverty and living in a metropolitan area increase murder rates.

It should be noted that there was an outlier in the data: the District of Columbia. This outlier could very well have skewed the data, and if this test were to be run again, it may be useful to either group it in with another state or eliminate it completely from the test. The other limitation is that since I used a frequentist approach, priors were ignored. This means that the results might look more powerful than they actually are. If this test were to be run again, it may be both useful and interesting to see the Bayesian version.